



ForgeGIS™

Deterministic GPU Geospatial for Air-Gapped Mission Environments

GPU-accelerated geospatial compute that deploys where classified work happens. A pure-Java Maven distribution with no first-party native code and no GDAL native bindings drops into existing JVM environments — air-gapped, accredited, or enterprise — without coordinating separate native binary distribution or shared-library signing for ForgeGIS itself. 239 operations, verified numerically deterministic across every invocation surface, so the audit and reproducibility guarantees regulated work requires hold whether an analyst, a script, or an AI agent makes the call.

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first-party native binaries to accredit

single shaded Java jar; no GDAL system install, no per-platform shared libraries

239

deterministic operations across 11 categories

byte-for-byte reproducible; verified each validation run

7.5×

geomean speedup vs. GDAL CLI

on a single commodity laptop; no server-class infrastructure required

Deploys into accredited environments

Because ForgeGIS contains no first-party native code and no GDAL native bindings, deployment to air-gapped or classified environments does not require coordinating separate native binary distribution, multi-source GLIBC validation, or shared-library signing for ForgeGIS itself. The required external components — JDK, GPU driver, OpenCL runtime — all have established procurement and approval paths. The deployment artifact is the same jar that runs unrestricted.

Determinism the mission can audit

Every shipped operation has a documented validation oracle, unit-test coverage, and benchmark coverage. Outputs are verified byte-for-byte reproducible across repeated runs. The MCP agent surface dispatches to the same compute implementations as the direct Java API — a parameter-marshaling shim, not a parallel path — so an agent-invoked result is identical to the scripted one. Correctness, surface-equivalence, and coverage are each independently checkable.

The line-of-sight surface, GPU-resident

Viewshed, batched multi-observer viewshed, intervisibility, Fresnel zone, horizon angle, insolation, sky-view factor, and shadow — the propagation and visibility operations that defense, IC, and aerospace workflows depend on, run on the GPU and composable into multi-step pipelines. Terrain, bathymetry, RF/visibility, and routing categories are first-class, not afterthoughts bolted onto a raster engine.

VALIDATED AGAINST REAL-WORLD TERRAIN

Real DEMs, real products, no fabrication

ForgeGIS operations are exercised end-to-end against real digital elevation datasets spanning high-relief and low-relief terrain — point elevations, viewsheds at observer height, slope and aspect rasters, elevation profiles, ray-cast intervisibility, and contour generation, producing real GeoTIFF products. Results are verified deterministic across repeated runs, and the validation harness re-executes each operation and compares outputs byte-for-byte. The numerical guarantees a scripted workflow depends on hold without exception when the same operations are invoked programmatically.

A VALIDATION FRAMEWORK AN EVALUATOR CAN INTERROGATE

Every operation has a documented oracle

Outputs are validated against domain-appropriate references rather than a single universal oracle: JTS for Cartesian predicates and topology, PostGIS for spheroidal computations on the WGS84 ellipsoid, GDAL for raster operations with direct equivalents, and analytic numerical fixtures with known closed-form outputs where no external oracle exists. A coverage matrix tracks every operation against oracle, MCP exposure, unit-test, and benchmark coverage.

Three questions, three defensible answers

The framework is built so an evaluator can ask — and get a defensible answer to — three independent questions: Is the operation correct? (oracle match), does it produce the same result however it is invoked? (surface-equivalence by construction), and has validation kept pace with the catalog? (the coverage matrix). That separation is what audit and reproducibility requirements in regulated work actually demand.

PERFORMANCE HIGHLIGHTS

7.5×

geomean across 225 head-to-head records vs. GDAL CLI — no category excluded, 206 wins (June 2026 run)

9–65×

multi-stage GPU-resident pipelines at 1024²–4096² rasters, vs. equivalent GDAL CLI chains

RTX 5070

entire validation run measured on a single commodity laptop — no server-class footprint

Honest by construction. The headline geomean is computed over every operation with a GDAL CLI equivalent — no category excluded and no cherry-picked subset. Small-raster cases where GDAL's process-spawn overhead dominates are reported but kept out of headline figures, and format-conversion round-trips are positioned at parity rather than claimed as wins. The full per-operation verdict record, both validation and product commit hashes, and the benchmark methodology ship with the Technical Brief.

New in 1.1

A non-linear DAG pipeline engine imports existing QGIS .model3 models and runs branching raster ↔ vector workflows GPU-resident; an expanded vector surface adds boolean overlay, dissolve, Voronoi, and variable-buffer operations; and a bundled EGM96 geoid makes orthometric/ellipsoidal vertical-datum conversion work out of the box — relevant wherever elevation reference frames matter to the product.

Request the Technical Brief and a live demonstration

The full Technical Brief includes the complete 239-operation catalog, validation oracles and methodology, per-category benchmarks, and air-gapped deployment guidance. An evaluation build is available to qualified programs on request.

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